# PLAN AND PREPARE THE PROCESS FOR THE WIRING AND CONNECTION,TESTING, INSPECTING,COMMISSIONING AND MAINTAINING ELECTRICAL INSTALLATION AND CONTROL SYSTEMS

## PRACTICAL MODULE

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|---|---|
| **Module #** | 671101000-PS-02 |
| **NQF Level** | level 3, |
| **Notional hours** | 50 |
| **Credit(s)** | Cr 5 |
| **Occupation or Specialisation** | 671101000: Occupational Certificate: Electrician |
| **Associated Occupation** | 671101: Electrician |

# INTRODUCTION

Congratulations on completing the program. As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

## Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility.** Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

# *Instructions to the Assessor*

Congratulations on your appointment as a assessor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

# *Provider Accreditation Requirements for the Module*

***Physical Requirements:***

Demonstrate access to:

Electrical environment

categories of tools, equipment and processes in order for learners to be able to plan worksite set up for installing, wiring and connecting electrical equipment and control systems

***Human Resource Requirements:***

Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, commissioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within en electrical-related industry with a recognized Trade Certificate.

Trainer/learner ratio 1 to 12

Workshop coach/learner ratio 1 to 4

***Legal Requirements:***

Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Partner (AQP)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Requirements

**Critical Practical Activities to be Assessed Externally for the Module**

None

**Exemptions**

# *PRACTICAL SKILL MODULE SPECIFICATIONS*

**List of Practical Skill Module Specifications**

671101000-PM-01 Use hand, power and measuring tools, L3, Cr28

671101000-PS-02 Plan worksite set up for installing, wiring and connecting Electrical equipment and control systems,L3, Cr5

671101000-PS-03 Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems,L3,Cr3

671101000-PS-04 Install wire-ways,L4, Cr5

671101000-PS-05 Install, wire and connect electrical equipment and control systems, L4,Cr38

671101000-PS-06 Conduct pre-commission inspection (power off) and test new and existing installations, L4,Cr5

671101000-PS-07 Carry out commissioning tests,L4,Cr13

671101000-PS-08 Fault find and repair electrical control systems and electrical installations, L4, Cr22

# *SECTION A-Guidelines for Practical Skills*

# *PM-02-PS01: Undertake a preliminary risk assessment*

***Scope of Practical Skill***

Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, the learner should be able to:

PA0101 Conduct a preliminary risk assessment.

PA0102 Identify the hazards.

PA0103 Relate the risk to the hazard.

PA0104 Conduct mitigation of the risk.

***Applied Knowledge***

AK0101 Safety procedures

AK0102 Statutory requirements for hazardous conditions/material/substance.

AK0103 Risk assessment procedure for specific context

AK0104 Inspection procedure

AK0105 Recording/reporting procedure

AK0106 Approval procedures

## OBSERVATION EVALUATION

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|---|---|
| **NAME:** .....<br>**SURNAME:** .....<br>**COMPANY**.....<br>**ID**..... | **EVALUATION CHECKLIST**<br><br>DATE:<br><br>TIME: |

| **EVALUATION CRITERIONS** | **MET REQUIREMENTS** | **DID NOT MEET REQUIREMENTS** | **COMMENTS OR ACTION REQUIRED** |
|---|---|---|---|
| | | | |
| **ABILITY OF THE LEARNER TO:** | | | |
| 1. Carry out Risk assessments, including work related hazards and ergonomics, and complete reports as per statutory requirements. | | | |
| 2. Observe Evidence of mitigation in accordance with standard operating procedures | | | |
| **3.** Complete approved | | | |

# *SECTION B-Guidelines for Practical Skills*

# *PM-02-PS02: Read and interpret electrical drawings*

***Scope of Practical Skill***

Given work instructions, work area, case study, documents, templates, the learner should be able to:

PA0201 Identify signs and symbols

PA0202 Read and interpret electrical drawings

***Applied Knowledge***

AK0201 Methods of interpreting scales

AK0202 Techniques of interpreting diagrams

AK0203 Procedures for evaluating the work site.

## OBSERVATION EVALUATION

| | |
|---|---|
| **NAME:** .....<br>**SURNAME:** .....<br>**COMPANY**.....<br>**ID**..... | **EVALUATION CHECKLIST**<br><br>DATE:<br><br>TIME: |

| **EVALUATION CRITERIONS** | **MET REQUIREMENTS** | **DID NOT MEET REQUIREMENTS** | **COMMENTS OR ACTION REQUIRED** |
|---|---|---|---|
| | | | |
| **ABILITY OF THE LEARNER TO:** | | | |
| 1. Correctly identify signs and symbols were according to the IEC( International Electrical compliance standards) | | | |
| 2. Correctly interpret electrical diagrams, using schematics, plans, wiring and layout representations according to worksite requirements. | | | |
| **GENERAL COMMENTS OF OBSERVER:** | | | |
| | | | |

# *SECTION C-Guidelines for Practical Skills*

# *PM-02-PS03: Plan the worksite set-up*

***Scope of Practical Skill***

*Given work instructions, work area, case study, documents, the learner should be able to:* PA0201 Compile a list of tools, material and equipment required

PA0202 Compile a list of human resources required

PA0203 Compile a work flow diagram

***Applied Knowledge***

AK0201 Knowledge of tools, material and equipment required to perform various functions within the worksite

AK0202 Knowledge of human resources required to perform various functions within the worksite

AK0203 Analytic techniques for interpreting work instructions

AK0204 Knowledge of workflow processes

## OBSERVATION EVALUATION

| NAME: .....<br>SURNAME: .....<br>COMPANY.....<br>ID..... | **EVALUATION CHECKLIST**<br><br>DATE:<br><br>TIME: |
|---|---|

| **EVALUATION CRITERIONS** | **MET REQUIREMENTS** | **DID NOT MEET REQUIREMENTS** | **COMMENTS OR ACTION REQUIRED** |
|---|---|---|---|
| | | | |
| **ABILITY OF THE LEARNER TO:** | | | |
| 1. List the relevant tools, material and equipment and record quantities in accordance with the task specification | | | |
| 2. List the correct human resources that meet the job specification | | | |
| 3. Analyse the work flow / work order correctly and apply in accordance with the task specification | | | |
| **GENERAL COMMENTS OF OBSERVER:** | | | |

| LOGBOOK | | | | |
|---|---|---|---|---|
| **Date** | **Assignment task** | **Start** | **Finish** | **Total Hours** |
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